

Final

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Github Repository

Set Up

```
library(tidyverse)
library(here)
library(janitor)

nests <- read.csv(here("data", "nest_data_final.csv"))
```

Problems

Problem 1. Research Writing

a. Transparent Statistical Methods

In part 1, they used a binomial “log odds” regression to relate and predict (*Recurvirostra americana*) habitat occupation to distance from water edge.

In part 2, they used a Chi-Squared test to compare expected vs observed counts for habitat type and occupation by species.

b. Suggestions for Rewriting

Utilizing binomial logistic regression analysis, we found a significant relationship between distance from the water’s edge and American Avocet (*Recurvirostra americana*) microhabitat use.

Utilizing a Chi-Squared test, we found a significant relationship between bird species (American Avocet, Forster’s Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further Analysis

One additional variable that could influence the probability of American Avocet (*Recurvirostra americana*) microhabitat use is **salinity**. This variable is continuous, numerical (ppm) and often calculated after measuring electrical conductivity in micro Siemens per centimeter ($\mu\text{S}/\text{cm}$). Salinity influences a plethora of biological interactions, specifically dehydration, [bodyweight](#) and [feeding/preening behavior in American Avocet chicks](#).

Problem 2. Data Analysis

a. Clean & Wrangle

```
nests_clean <- nests |>
  select(case_control, ant_species, height_cm) |>
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  mutate(ant_name = case_match(
    ant_species,
    "AB" ~ "C.sjostedti",
    "RRB" ~ "C.mimosae",
    "BBR" ~ "C.nigriceps"
  )) |>
  mutate(ant_name = as_factor(ant_name),
    ant_name = fct_relevel(ant_name,
                          "C.sjostedti",
                          "C.mimosae",
                          "C.nigriceps"))

str(nests_clean)
```

```
'data.frame': 270 obs. of 4 variables:
 $ case_control: int 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species : chr "RRB" "RRB" "RRB" "RRB" ...
 $ height_cm : num 392 310 513 154 324 ...
 $ ant_name : Factor w/ 3 levels "C.sjostedti",...: 2 2 2 2 1 2 3 2 1 3 ...
```

```
slice_sample(nests_clean, n = 10)
```

```
  case_control ant_species height_cm ant_name
1            0          BBR      54.0 C.nigriceps
```

2	0	RRB	200.5	C.mimosae
3	0	RRB	114.5	C.mimosae
4	0	RRB	279.0	C.mimosae
5	1	RRB	204.5	C.mimosae
6	0	RRB	204.0	C.mimosae
7	0	RRB	329.0	C.mimosae
8	0	RRB	236.0	C.mimosae
9	0	RRB	128.0	C.mimosae
10	1	RRB	525.0	C.mimosae

b. Response Variable

The 1s and 0s present within the “case_control” column of the data represent if a bird nest was observed (1) or not (0) within a specific tree.

c. Purpose of Study

Tree height is a continuous, numerical variable measured in centimeters. As tree height increases, the probability of nest occurrence would increase because the tree contains more space for nesting and is further away from the ground and therefore potential predators. (Section 3: Results, ¶2)

Acacia ant species is a categorical, nominal variable. The four studied ant species differ in aggression of protecting their host trees, as aggressive protection increases, bird nesting also increases because of this inherent nest protection. (Section 3: Results, ¶3)

d. Table of Models

Model number	Model Description
1	Ants and Height: No Interaction
2	Ants and Height: With Interaction

Model number	Predictor List	Interaction Term?
1	Ant Species, Tree Height	No Interaction
2	Ant Species, Tree Height	With Interaction

e. Running the Models

```
model1 <- glm(case_control ~ ant_name + height_cm,  
              data = nests_clean,  
              family = "binomial")  
model1
```

```
Call: glm(formula = case_control ~ ant_name + height_cm, family = "binomial",  
          data = nests_clean)
```

Coefficients:

(Intercept)	ant_nameC.mimosae	ant_nameC.nigriceps
-4.950302	1.717100	2.797757
height_cm		
0.008018		

Degrees of Freedom: 269 Total (i.e. Null); 266 Residual

Null Deviance: 286

Residual Deviance: 250.7 AIC: 258.7

```
model2 <- glm(case_control ~ ant_name*height_cm,  
              data = nests_clean,  
              family = "binomial")  
model2
```

```
Call: glm(formula = case_control ~ ant_name * height_cm, family = "binomial",  
          data = nests_clean)
```

Coefficients:

(Intercept)	ant_nameC.mimosae
-3.693410	0.848334
ant_nameC.nigriceps	height_cm
-12.279738	0.003895
ant_nameC.mimosae:height_cm	ant_nameC.nigriceps:height_cm
0.002597	0.084541

Degrees of Freedom: 269 Total (i.e. Null); 264 Residual

Null Deviance: 286

Residual Deviance: 225.8 AIC: 237.8

f. Selecting the Best Model

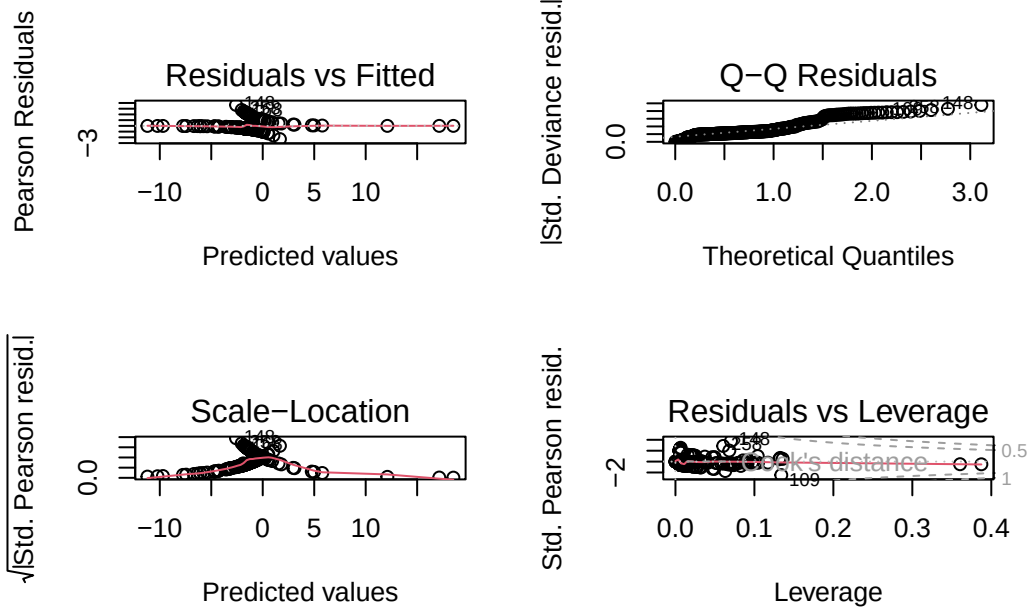
```
AIC(model1, model2)
```

	df	AIC
model1	4	258.7430
model2	6	237.8042

The best model as determined by Akaike's Information Criterion (AIC) was the model predicting the probability of bird nest occurrence as predicted by ant species interacting with tree height.

g. Checking Diagnostics

```
par(mfrow = c(2,2))  
plot(model2)
```



h. Visualizing Model Predictions

i. Writing a Caption for Figure

j. Calculating Model Predictions

k. Interpreting Results

Problem 3. Affective & Exploratory Visualizations

a. Comparing Visualizations

b. Designing an Analysis

c. Checking Assumptions & Running Analysis

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g. Sharing Affective Visualization